

CS 255 Business Requirements Document Template

Vonds Auguste

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design and implement the DriverPass system to enhance the success rate of individuals passing their driving tests.
- The client, DriverPass, wants a system that integrates online classes, practice tests, and scheduling of on-the-road driving sessions. The system should be easy for customers to use and should facilitate internal operations (such as user account management and scheduling) for the client's team.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants the system to address the gap in effective driver training by offering an online platform that allows customers to access class materials, take practice tests, schedule driving sessions, and interact with instructors or administrators.
- The problem being addressed is the inefficient and fragmented process of preparing and scheduling driving tests.
- System components needed include:
 - A web-based user interface for students and internal staff
 - A backend database to store user information, session schedules, and test results

- Integration with external DMV systems to keep training materials and policies up to date
- Authentication and authorization components for secure access

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Provide a user-friendly interface for customers to create and manage their accounts, book or cancel driving sessions, and access study materials.
- Allow administrators, IT officers, drivers, and secretaries to have role-based access to the system for performing tasks such as resetting passwords, updating schedules, and managing user accounts.
- Ensure system security, reliability, and scalability to support future feature enhancements.
- Improve the pass rate of customers by offering comprehensive training materials and convenient practice scheduling.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system should be web-based and hosted on a reliable cloud platform for high availability.
- It should load pages within two seconds under normal usage conditions.
- The system should synchronize updates (e.g., from the DMV) at least once per day to remain current.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should run on common operating systems (Windows, macOS, Linux) and modern web browsers (Chrome, Safari, Edge).
- A backend relational database (e.g., PostgreSQL) is required to store user, session, and test data.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system must enforce role-based access to ensure only authorized personnel can perform certain tasks.
- All actions should be timestamped and logged for auditing.
- Input should not be case-sensitive where not required (e.g., usernames are not case-sensitive, but passwords are).

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Administrators should be able to add, remove, or modify user access and package offerings without changing the underlying code.
- The system design should allow for easy updates when the underlying technologies (e.g., web frameworks, database versions) change.
- IT administrators must have sufficient privileges to implement changes, updates, and patches as needed.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Users must log in with a username and password; two-factor authentication (2FA) is recommended.

- The connection between the client and server must be secured using HTTPS protocols.
- After five failed login attempts, an account should be temporarily locked to deter brute-force attacks.
- If a user forgets their password, they should be able to reset it via an email-based verification process.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall validate user credentials when logging in.
- The system shall allow customers to schedule, reschedule, or cancel driving sessions online.
- The system shall record and log all changes made by users for auditing purposes.
- The system shall allow administrators to reset passwords and modify user accounts.
- The system shall generate detailed activity and transaction reports for internal analysis.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface must be intuitive and responsive, accessible via desktop and mobile browsers.
- Customers must be able to register, view training materials, take practice tests, and schedule driving sessions easily.
- Administrators need interfaces to view activity logs, reset passwords, and manage user roles.
- Secretaries and drivers should be able to manage session schedules and update student records.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- All users have consistent internet access.
- DMV updates will be provided in a standard, compatible format.
- Users have basic computer literacy.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Development and integration are constrained by time and budget as initially defined by DriverPass.
- Advanced features beyond the core requirements may be deferred to future development phases.
- Hardware and bandwidth resources may limit system performance under peak usage.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

DriverPass Project Gantt Chart

